



Visitor feedback, heard in the host's own language.

SabiEcho is an offline app for community tourism hosts. A visitor's review in **English or French**, typed, spoken or shared from WhatsApp, is turned into **27 numbered meanings** and played back in the host's language as **phrases recorded by real people**. The pilot language is **Fon (Benin)**.

• The problem

Many community tourism hosts do not read English or French well, so feedback that could improve their visits goes unheard. Machine translation into languages like Fon is rare and unreliable, and rural connections are patchy.

• How it works

- 1 Capture**
Type a review, record the visitor, upload a voice note or share it from WhatsApp.
- 2 Understand**
Speech is transcribed on the phone, then each part of the review is matched to one of 27 meanings, such as "the price was too high".
- 3 Hear**
Each meaning plays as a recording in the host's language. If unsure, the app says so and keeps the review for a person.

• Example

"We had a great time. The host explained everything clearly, the food was excellent, but the farm was hard to find."

- 01** Xome hun jonɔ ɔ
The visitor was happy
- 04** Tinme towe le nyó nukún yetɔn me
They liked your explanations
- 14** Ye ɖu nú e ye ɖu bo nu le é sín viví
They liked what they ate and drank
- 25** É vewũ nú ye bo ye na mɔ ten ɔ
They had difficulty finding the place

• What hosts get

- Instant meaning, in their language.** Every review is saved with what it means and the exact words it came from.
- Insights.** How visitors felt over 7 days, 30 days or all time, whether it is improving, what they love and what to fix first, with practical advice, all playable in Fon.
- Phrasebook.** All 27 meanings with their recordings, to learn and replay.
- Works offline.** After one visit online, the app, the AI and the chosen language's recordings work without signal.
- Private by design.** Reviews stay on the phone. There is no account and no server.

• Improved by local language reviewers

Trusted reviewers sign in with a personal code. They correct reviews the app got wrong, teach it new phrases, and record and check voice clips. The phone retrains its classifier in seconds, offline, and corrections can be shared between devices as a file.

• How it is built

An installable web app (PWA) running entirely in the browser. Whisper transcribes speech; a multilingual sentence model turns each part of a review into numbers; a small classifier trained on 604 English and French examples maps them to the 27 meanings. Because the output is a number, **adding a language means recording 26 phrases, not retraining**.

• Status and next steps

Live and pilot-ready, with 26 of 27 Fon phrases recorded. Next: a pilot with hosts in Benin, native-speaker review of three newer Fon phrases, and language reviewers for the next languages.

27

numbered meanings cover overall feeling and 11 topics

0.79

accuracy (F1) on unseen reviews, vs 0.34 for a small LLM

0 MB

of data needed after the first visit

~7 s

to retrain on the phone after a correction

15

African languages ready to receive recordings